



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)

Accredited by NAAC with 'A+' Grade.

Recognised as Scientific and Industrial Research Organisation

SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., INDIA

Regulation: R23		II / IV - B.Tech. I - Semester							
ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING									
COURSE STRUCTURE (With effect from 2023-24 admitted Batch onwards)									
Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
B23BS2101	Discrete Mathematics and Graph Theory	BS	3	0	0	3	30	70	100
B23HS2101	Universal Human Values -II: Understanding Harmony and Ethical Human Conduct	HS	2	1	0	3	30	70	100
B23AM2101	Artificial Intelligence	PC	3	0	0	3	30	70	100
B23CS2102	Database Management Systems	PC	3	0	0	3	30	70	100
B23CS2103	Object Oriented Programing Through Java	PC	3	0	0	3	30	70	100
B23CS2104	Database Management Systems Lab	PC	0	0	3	1.5	30	70	100
B23CS2105	Object Oriented Programing through Java Lab	PC	0	0	3	1.5	30	70	100
B23CS2106	Python Programming	SEC	0	1	2	2	30	70	100
B23MC2101	English Proficiency	MC	2	0	0	-	30	-	30
TOTAL			16	2	8	20	270	560	830

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23BS2101	BS	3	--	--	3	30	70	3 Hrs.
DISCRETE MATHEMATICS AND GRAPH THEORY								
(Common to CSE, CSBS, AIML, IT, AIDS, CSG, CIC, CSIT)								
Course Objectives: Students are expected to								
1.	Understand the mathematical arguments using logical connectives and quantifiers and verify the validity of arguments using propositional, predicate logic and truth tables.							
2.	Understand various types of relations and discuss various properties of the relations							
3.	Know about the concepts of counting techniques and how to solve the recurrence relations.							
4.	Understand the concepts in graphs and trees.							
Course Outcomes: At the end of the course, Students will be able to								
S.No	Outcome							Knowledge Level
1.	Use the concepts of propositional and predicate logic to verify the arguments for their validity.							K3
2.	Apply the knowledge of set theory to understand relations, functions and their properties.							K3
3.	Solve different counting problems and recurrence relations.							K3
4.	Use the concepts of graphs and their representations.							K3
5.	Determine different multi graphs and tree structures.							K3
SYLLABUS								
UNIT-I (10Hrs)	Mathematical Logic: Propositional Calculus: Statements and Notations, Connectives, Well Formed Formulas, Truth Tables, Tautologies, Equivalence of Formulas, Duality Law, Tautological Implications, Normal Forms, Theory of Inference for Statement Calculus, Consistency of Premises, Indirect Method of Proof, Predicate Calculus: Predicates, Predicative Logic, Statement Functions, Variables and Quantifiers, Free and Bound Variables, Inference Theory for Predicate Calculus.							
UNIT-II (10 Hrs)	Set Theory: Sets: Operations on Sets, Principle of Inclusion-Exclusion, Relations: Properties, Operations, Partition and Covering, Transitive Closure, Equivalence, Compatibility and Partial Ordering, Hasse Diagrams, Lattice and its Properties. Functions: Bijective, Composition, Inverse, Permutation, and Recursive Functions.							
UNIT-III (12Hrs)	Combinatorics and Recurrence Relations: Basis of Counting, Permutations, Permutations with repetitions, Circular and Restricted Permutations, Combinations,							

	Restricted Combinations, Binomial and Multinomial Coefficients and Theorems. Recurrence Relations: Generating Functions, Function of Sequences, Partial Fractions, Calculating Coefficient of Generating Functions, Recurrence Relations, Formulation as Recurrence Relations, Solving Recurrence Relations by Substitution and Generating Functions, Method of Characteristic Roots, Solving Inhomogeneous Recurrence Relations
UNIT-IV (10 Hrs)	Graph Theory: Basic Concepts, Graph Theory and its Applications, Subgraphs, Graph Representations: Adjacency and Incidence Matrices, Isomorphic Graphs, Paths and Circuits, Eulerian and Hamiltonian Graphs.
UNIT-V (08Hrs)	Multi Graphs: Multi graphs, Bipartite and Planar Graphs, Euler's Theorem, Graph Coloring and Covering, Chromatic Number, Trees and their properties, Spanning Trees- BFS and DFS Spanning Trees, Prim's and Kruskal's Algorithms.
Textbooks:	
1.	Discrete Mathematical Structures with Applications to Computer Science, J. P. Tremblay and P. Manohar, Tata McGraw Hill.
2.	Discrete Mathematics for Computer Scientists and Mathematicians, J. L. Mott, A. Kandel and T. P. Baker, 2nd Edition, Prentice Hall of India.
Reference Books:	
1.	Elements of Discrete Mathematics-A Computer Oriented Approach, C. L.Liu and D. P. Mohapatra, 3rd Edition, Tata McGraw Hill.
2.	Theory and Problems of Discrete Mathematics, Schaum's Outline Series, Seymour Lipschutz and Marc Lars Lipson, 3rd Edition, McGraw Hill.
3.	Discrete Mathematical Structures, Bernand Kolman, Robert C. Busby and Sharon Cutler Ross, PHI.
4.	Discrete Mathematics, S. K. Chakraborty and B.K. Sarkar, Oxford, 2011.
5.	Discrete Mathematics and its Applications with Combinatorics and Graph Theory, K. H. Rosen, 7th Edition, Tata McGraw Hill.
e-Resources :	
1.	https://nptel.ac.in/courses/106105192
2.	https://archive.nptel.ac.in/courses/111/106/111106102/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23HS2101	HS	2	1	--	3	30	70	3 Hrs.
UNIVERSAL HUMAN VALUES-II: UNDERSTANDING HARMONY AND ETHICAL HUMAN CONDUCT								
(Common to all Programmes of Engineering)								
Course Objectives: The objective of this course is to make the student aware of								
1	Essential complementarity between 'Values' and 'Skills' to ensure sustained happiness and prosperity which are the core aspirations of all human beings.							
2	Harmony in the human being, family, society and nature/existence							
3	Holistic perspective towards life, profession and happiness							
Course Outcomes: At the end of this course student will be able to								
S. No.	Outcome							Knowledge Level
1	Explain the role of value education in achieving basic human aspirations.							K2
2	Summarize needs to obtain harmony in self(I).							K2
3	Describe criteria for human-human relationship and harmony in society							K2
4	Explain four orders of nature and our existence							K2
5	Interpret significance of harmony in holistic development							K2
SYLLABUS								
UNIT-I (9 Hrs)	Introduction to Value Education: Understanding Value Education- Need, Basic Guidelines, Content and Process for Value Education Purpose and motivation for the course. Self-exploration as the Process for Value Education - Sharing about Oneself. Myers-Briggs Type Indicator (MBTI) Personality Test. Continuous Happiness and Prosperity – the Basic Human Aspirations and their Fulfilment. Right Understanding, Relationship and Physical Facility (Holistic Development and the Role of Education) - Exploring Human Consciousness. Happiness and Prosperity – Current Scenario. Method to Fulfil the Basic Human Aspirations - Exploring Natural Acceptance-understanding and living in harmony at various levels.							
UNIT-II (9 Hrs)	Harmony in the Human Being: Understanding Human being as the Co-existence of the Self and the Body. Distinguishing between the Needs of the Self and the Body - Exploring the difference of Needs of Self (I) and Body (Happiness and Physical Facility). The Body as an Instrument of the Self (I)’ (I being the doer, seer and enjoyer). Understanding Harmony in the Self(I) - Exploring Sources of Imagination in the Self(I). Harmony of the Self (I) with the Body (characteristics and activities of ‘I’ and harmony							

	in 'I'). Programme to ensure self-regulation(<i>Sanyam</i>) and Health(<i>Swasth</i>)- Exploring Harmony of Self (I) with the Body.
UNIT-III (9 Hrs)	Harmony in the Family and Society: Harmony in the Family – the Basic Unit of Human Interaction. 'Trust' – the Foundational Value in Relationship- Exploring the Feeling of Trust- intention and competence. 'Respect' – as the Right Evaluation- Exploring the Feeling of Respect. Other Feelings, Justice in Human-to-Human Relationship. Understanding Harmony in the Society- (society being an extension of family). Vision for the Universal Human Order- Exploring Systems to fulfil Human Goal.
UNIT-IV (6 Hrs)	Harmony in the Nature/Existence: Understanding the harmony in the Nature. Interconnectedness and mutual fulfillment among the four orders of nature- recyclability and self-regulation in nature. Realizing Existence as Co-existence at All Levels - Understanding Existence as Co-existence of mutually interacting units in all pervasive space. Holistic perception of harmony at all levels of existence.
UNIT-V (9 Hrs)	Implications of the Holistic Understanding – a Look at Professional Ethics: Natural acceptance of human values. Definitiveness of (ethical) human conduct. A Basis for Humanistic Education, Humanistic Constitution and Humanistic Universal Order. Competence in professional ethics: a. Ability to utilize the professional competence for augmenting universal human order b. Ability to identify the scope and characteristics of people friendly and eco-friendly production systems, c. Ability to identify and develop appropriate technologies and management patterns for above production systems. Holistic technologies, production systems and management models- typical case studies. Strategies for transition towards value based life and profession (from the present state to Universal Human Order): a. At the level of individual: as socially and ecologically responsible engineers, technologists and managers, b. At the level of society: as mutually enriching institutions and organizations.
Text Books	
1.	R R Gaur, R Sangal, G P Bagaria. "Human Values and Professional Ethics", Excel Books, New Delhi, 2010
2.	R R Gaur, R Asthana, G P Bagaria. "Teachers' Manual for A Foundation Course in Human Values and Professional Ethics", 2nd Revised Edition, Excel Books, New Delhi, 2019. ISBN 978-93-87034-53-2
Reference Books:	
1.	Jeevan Vidya: Ek Parichaya, A Nagaraj, Jeevan Vidya Prakashan, Amarkantak, 1999.
2.	Human Values, A.N. Tripathi, New Age Intl. Publishers, New Delhi, 2004.
3.	The Story of Stuff (Book).

4.	The Story of My Experiments with Truth - by Mohandas Karamchand Gandhi
5.	Small is Beautiful - E. F Schumacher
6.	Slow is Beautiful - Cecile Andrews
7.	Economy of Permanence - J C Kumarappa
8.	Bharat Mein Angreji Raj – Pandit Sunderlal
9.	Rediscovering India - by Dharampal
10.	Hind Swaraj or Indian Home Rule - by Mohandas K. Gandhi
11.	India Wins Freedom - Maulana Abdul Kalam Azad
12.	Vivekananda - Romain Rolland (English)

e-Resources

1.	https://fdp-si.aicte-india.org/UHV%20II%20Teaching%20Material/UHV%20II%20Lecture%2023-25%20Ethics%20v1.pdf
2.	https://fdp-si.aicte-india.org/UHV-II%20Class%20Note.php
3.	https://fdp-si.aicte-india.org/download/FDPTeachingMaterial/3-days%20FDP-SI%20UHV%20Teaching%20Material/Day%203%20Handouts/UHV%203D%20D3-S2A%20Und%20Nature-Existence.pdf



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B23AM2101	PC	3	--	--	3	30	70	3 Hrs.
ARTIFICIAL INTELLIGENCE								
(For AIML)								
Course Objectives: The main objectives of the course is to								
1.	The student should be made to study the concepts of Artificial Intelligence.							
2.	The student should be made to learn the methods of solving problems using Artificial Intelligence.							
3.	The student should be made to introduce the concepts of Expert Systems.							
4.	To understand the applications of AI, namely game playing, theorem proving, and machine learning.							
5	To learn different knowledge representation techniques							
Course Outcomes: At the end of the course Students will be able to,								
S.No	Outcome							Knowledge Level
1.	Use state space representation for solving AI problems.							K3
2.	Apply Heuristic search techniques to solve AI problems.							K3
3.	Apply complex knowledge representation methods including predicate logic, semantic networks, frames and inheritance, conceptual dependency, scripts, constraint propagation to the knowledge in different domains.							K3
4.	Apply various inference techniques on the given facts.							K3
5.	Apply principles and techniques from machine learning and expert systems to solve real-world problems.							K3
SYLLABUS								
UNIT-I (08 Hrs)	Introduction: Introduction to AI, Foundation of AI and history of AI, Domains of AI, State Space Representation of AI Problems (Water Jug Problem, 8 Puzzle Problem, TSP), Problem characteristics of AI, intelligent agents: Agents and Environments, structure of agents, problem solving agents.							
UNIT-II (10 Hrs)	Searching: Searching for solutions, uniformed search strategies – Breadth first search, depth first Search. Search with partial information (Heuristic search) Hill climbing, A*, AO* Algorithms, applications of Hill climbing and A* algorithms to solve AI Problems.							

UNIT-III (12 Hrs)	Representation of Knowledge: Knowledge representation issues, predicate logic- logic programming, semantic nets- frames and inheritance, Conceptual dependency and Scripts, constraint propagation, representing knowledge using rules, rules based deduction systems. Reasoning under uncertainty.
UNIT-IV (10 Hrs)	Logic concepts: First order Predicate logic. Inference in first order logic, propositional vs. first order inference, Conversion of WFF to Clause form, Unification Algorithm, forward chaining, Backward chaining, Resolution Algorithm. Game Playing: Adversarial search, Games, mini-max algorithm, optimal decisions in multiplayer games, Problem in Game playing, Alpha-Beta pruning.
UNIT-V (10 Hrs)	Introduction to Machine learning: learning from observation Inductive learning, Decision trees, Explanation based learning, Statistical Learning methods, Reinforcement Learning. Expert Systems: Introduction to Expert Systems, Advantages and disadvantages and applications of Expert systems, Architecture of expert systems, Roles of expert systems –Knowledge Acquisition, Meta knowledge Heuristics.
Textbooks:	
1.	S. Russel and P. Norvig, “Artificial Intelligence – A Modern Approach”, Second Edition, Pearson Education (2005).
2.	Kevin Night and Elaine Rich, Nair B., “Artificial Intelligence (SIE)”, Third Edition, Mc Graw-Hill (2016).
Reference Books:	
1.	David Poole, Alan Mackworth, Randy Goebel,” Computational Intelligence: a logical approach”, Oxford University Press (1998).
2.	G. Luger, “Artificial Intelligence: Structures and Strategies for complex problem solving”, Fifth Edition, Pearson Education (2009).
3.	J. Nilsson, “Artificial Intelligence: A new Synthesis”, Elsevier Publishers (2010).
4.	Saroj Kaushik, “Artificial Intelligence”, Second Edition, CENGAGE Learning (2011).
e-Resources	
1.	https://ai.google/
2.	https://swayam.gov.in/nd1_noc19_me71/preview

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CS2102	PC	3	--	--	3	30	70	3 Hrs.
DATABASE MANAGEMENT SYSTEMS								
(Common to CSE, AIML, CSIT & CSG Branches)								
Course Objectives: The main objectives of the course is to								
1.	Introduce database management systems							
2.	Analyze database through systematic database design approaches							
3.	Use SQL as a universal Database language							
4.	Demonstrate normalization							
5	Explain transaction management techniques							
Course Outcomes: At the end of the course Students will be able to,								
S.No	Outcome							Knowledge Level
1.	Describe database management systems fundamental concepts							K2
2.	Analyze databases using Conceptual and Logical database design							K4
3.	Apply SQL to Create, maintain and manipulate a relational database							K3
4.	Apply normalization for refining database schema							K3
5.	Illustrate Transaction management techniques.							K2
SYLLABUS								
UNIT-I (08 Hrs)	Introduction: Databases and Database Management Systems, Characteristics of DBMS, DBMS Vs File System, Database Users, Database applications. Brief introduction of different Data Models, Introduction to Relational Database Management Systems, Concepts of Schema, Instance, three tier schema architecture for data independence, Database system structure, Centralized and Client Server architecture for the database.							
UNIT-II (10 Hrs)	Relational Model: Introduction to relational model, concepts of domain, attribute, tuple, relation, importance of null values, constraints (Domain, Key constraints, integrity constraints) and their importance, Relational Algebra (select and project) . Entity Relationship Model: Introduction, Representation of entities, attributes, entity set, relationship, relationship set, constraints, extended features of ER model, conversion of ER diagrams to tables.							
UNIT-III (12 Hrs)	SQL: Simple Database schema, data types, table definitions (create, alter), Creating tables with relationship, implementation of key and integrity constraints, different DML operations (insert, delete, update), Basic SQL querying (select and project) using where clause, arithmetic & logical operations, SQL functions (Date and Time, Numeric, String							

	conversion). nested queries, sub queries, grouping, aggregation, ordering, relational set operations, implementation of different types of joins, view (updatable and non-updatable).
UNIT-IV (10 Hrs)	Schema Refinement (Normalization): Purpose of Normalization or schema refinement, concept of functional dependency, closure of functional dependencies, normal forms based on functional dependencies, 1NF, 2NF and 3 NF, Boyce-Codd normal form (BCNF), Lossless join and dependency preserving decomposition, MVD, Fourth normal form(4NF), Fifth Normal Form (5NF).
UNIT-V (10 Hrs)	Transaction Concept: Transaction State, ACID properties, Concurrent Execution of transactions, Schedules, Serializability, Recoverability, Testing for Serializability, Lock based and timestamp-based concurrency protocols, Implementation of Isolation, Failure Classification, ARIES Recovery algorithm, Introduction to Indexing Techniques, B+ Trees, operations on B+ Trees, Hash Based Indexing.
Textbooks:	
1.	Abraham Silberschatz, Henry F. Korth and S. Sudarshan (Author), Database System Concepts, 7th Edition, TMH, 2021.
2.	Raghu Ramakrishnan and Johannes Gehrke, Database Management Systems, 3rd Edition, Pearson, 2014
Reference Books:	
1.	C.J. Date, A. Kannan and S. Swamy Nathan, An Introduction to Database Systems, 8th Edition, Pearson, 2006.
2.	Elmasri Ramez and Navathe Shamkant, Fundamentals of Database System, 7th Edition, Pearson, 2017.
3.	Corlos Coronel, Steven Morris, Peter Robb, Database Principles Fundamentals of Design Implementation and Management, CBS publishers and Distributors, 2014.
e-Resources	
1.	https://nptel.ac.in/courses/106/105/106105175/
2.	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01275806667282022456_shared/overview

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CS2103	PC	3	--	--	3	30	70	3 Hrs.
OBJECT ORIENTED PROGRAMMING THROUGH JAVA								
(Common to CSE, CSG, CSIT, AIML & CIC Branches)								
Course Objectives:								
1.	To identify Java language components and how they work together in applications.							
2.	To learn the fundamentals of object-oriented programming in Java, including defining classes, invoking methods, using class libraries and collections.							
3.	To learn how to extend Java classes with inheritance and dynamic binding and how to use exception handling, file I/O in Java applications.							
4.	To understand how to design applications with threads in Java.							
5.	To understand how to use JDBC APIs for program development.							
Course Outcomes: At the end of the course Students will be able to								
S.No	Outcome							Knowledge Level
1.	Apply basic Object-Oriented Programming Concepts in Java.							K3
2.	Use Arrays, Collections and Strings to solve real world problems.							K3
3.	Apply the concepts of Inheritance, Interfaces and Packages for developing reusable programs.							K3
4.	Apply the concepts of Multithreading and Exception Handling to build an efficient and error free code.							K3
5.	Use Input/Output Streams and JDBC to manage data effectively.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Introduction to OOP: Basic concepts of OOP, Differences between Procedural and Object-Oriented Programming, Advantages of OOP, Applications of OOP.							
	Introduction to JAVA: Structure of JAVA, Features of JAVA, Data Types, JAVA Tokens, Control Statements.							
	Classes & Objects: Introduction, Class Declaration, Class Members, Declaration of Class Objects, Access Control for Class Members, Static Variables and Methods, User input to programs, Command Line Arguments, Method overloading.							
	Constructors: Default Constructor, Parameterized Constructor, Copy Constructor and Constructor Overloading, This Keyword.							
UNIT-II (10 Hrs)	Arrays: Introduction, Declaration and Initialization of Arrays, Storage of Array in Computer Memory, Accessing Elements of Arrays, Two-dimensional Arrays and Variable Size Arrays.							
	Collections: Array List, HashMap and HashSet							

	String Handling in Java: Introduction, Methods in String class, String Constant Pool and String Buffer class, Wrapper classes, Type Conversion.
UNIT-III (10 Hrs)	Inheritance: Introduction, Single inheritance, Multi-level inheritance, Hierarchical Inheritance, Method Overriding, Super Keyword, Final Keyword and Abstract Classes. Interfaces: Introduction, Declaration of Interface, Implementing Multiple Inheritance, Extending interfaces, Adapter classes. Packages: Introduction, Defining Package, Java util Classes and Interfaces, Java lang Classes, Importing Packages, Sub Packages, Access Modifiers.
UNIT-IV (10 Hrs)	Exception Handling: Introduction, Hierarchy of Standard Exception Classes, Keywords try, catch, throw, throws and finally Blocks, Multiple Catch Statements, Custom Exceptions, Nested try and catch Blocks. Multithreaded Programming: Introduction, Thread Life Cycle, Extending Thread class, Implementing Runnable interface, Thread Priorities, Thread Synchronization.
UNIT-V (10 Hrs)	File IO: Introduction, Hierarchy of Stream classes, Byte Streams, Character streams. Java Database Connectivity: Introduction, Structure of JDBC, JDBC Architecture, Types of JDBC Drivers, JAVA Database connection program for MySQL.
Textbooks:	
1.	The complete Reference Java, 12th edition (2022), Herbert Schildt, Publisher: McGraw Hill.
2.	JdbcApi Tutorial and Reference 3E (2003), by Maydene, Jon Ellis (Author), Jonathan Bruce (Author), Publisher: Addison-Wesley Professional.
3.	Joy with JAVA, Fundamentals of Object Oriented Programming, Debasis Samanta, Monalisa Sarma, Cambridge, 2023.
Reference Books:	
1.	Introduction to java programming, 9th edition (2014) by Y Daniel Liang, Publisher: Pearson.
2.	Murach's Java Programming, 5th edition (2017) Joel Murach, Publisher: Mike Murach.
3.	JAVA one step ahead, 1st edition (2017) Anitha Seth, B.L. Juneja, Oxford.
4.	Java: A Beginner's Guide, Eighth Edition 8th Edition (2018) by Herbert Schildt, Publisher: McGraw-Hill Education.
5.	Head First Java 3e (2021) (A Brain Friendly Guide) by Kathy Sierra & Bert bates, Publisher: O'Reilly.
6.	Programming with Java: A Primer 6E (2019) By Balagurusamy, Publisher: TMH.
e-Resources	
1.	https://nptel.ac.in/courses/106/105/106105191/
2.	https://www.coursera.org/learn/java-introduction
3.	https://docs.oracle.com/javase/tutorial/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CS2104	PC	--	--	3	1.5	30	70	3 Hrs.
DATABASE MANAGEMENT SYSTEMS LAB								
(Common to CSE, AIML, CSIT & CSG Branches)								
Course Objectives: The main objectives of the course is to								
1	Query a database using SQL DDL/DML Commands							
2	Use PL/SQL to implement procedures, functions, cursors and triggers							
3	Implement a DBMS mini project							
Course Outcomes: At the end of the course Students will be able to,								
S.No	Outcome							Knowledge Level
1	Apply SQL Commands for defining, constructing and manipulating databases							K3
2	Demonstrate PL/SQL to create Procedures, Functions, Cursors, and Triggers.							K3
3	Apply database design concepts for a real time problem							K3
SYLLABUS								
1	Creating, altering and dropping tables and inserting rows into a table (use constraints while creating tables) examples using SELECT command.							
2	Queries (along with sub Queries) using ANY, ALL, IN, EXISTS, NOT EXISTS, UNION, INTERSET, Constraints. Example: - Select the roll number and name of the student who secured fourth rank in the class.							
3	Queries using Aggregate functions (COUNT, SUM, AVG, MAX and MIN), GROUP BY, HAVING and Creation and dropping of Views.							
4	Queries using Conversion functions (to_char, to_number and to_date), string functions (Concatenation, lpad, rpad, ltrim, rtrim, lower, upper, initcap, length, substr and instr), date functions (Sysdate, next_day, add_months, last_day, months_between, least, greatest, trunc, round, to_char, to_date)							
5	i. Create a simple PL/SQL program which includes declaration section, executable section and exception –Handling section (Ex. Student marks can be selected from the table and printed for those who secured first class and an exception can be raised if no records were found) ii. Insert data into the student table and use COMMIT, ROLLBACK and SAVEPOINT in PL/SQL block.							
6	Develop a program that includes the features NESTED IF, CASE and CASE expression. The program can be extended using the NULLIF and COALESCE functions.							
7	Program development using WHILE LOOPS, numeric FOR LOOPS, nested loops using ERROR Handling, BUILT –IN Exceptions, USE defined Exceptions, RAISE- APPLICATION ERROR.							

8	Develop programs using creation of procedures, passing parameters IN and OUT of PROCEDURES.
9	Develop programs using creation of functions, invoke functions in SQL Statements and write complex functions.
10	Develop programs using a CURSOR, FOR UPDATE CURSOR, WHERE CURRENT of clause and CURSOR variables.
11	Develop Programs using BEFORE and AFTER Triggers, Row and Statement level Triggers
12	Create a table and perform the search operation on the tables using indexing and non-indexing techniques.
13	Mini Project Applying the concepts learnt in the lab.
Add on Programs:	
1	Connect a database using JDBC from a JAVA program.
2	Write a java program using JDBC connection to manipulate data in a database.
3	Create and manipulate a Database using MySQL.
Reference Books:	
1	Bob Bryla and Kevin Loney, Oracle Database 12c The Complete Reference (Oracle Press), McGraw-Hill Education, 2013.
2	Nilesh Shah, Database systems using oracle, 2nd Edition, Pearson Education India, 2016.
3	Der Lans, Van, Introduction To Sql: Mastering The Relational Database Language, 4th Edition (With Cd), Pearson Education India, 2007.
e-Resources	
1.	https://livesql.oracle.com/apex/f?p=590:1000

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CS2105	PC	--	--	3	1.5	30	70	3 Hrs.
OBJECT ORIENTED PROGRAMMING THROUGH JAVA LAB								
(Common to CSE, CSG, CSIT, AIML & CIC Branches)								
Course Objectives:								
1	Practice object-oriented programming in the Java programming language							
2	Implement Classes, Objects, Methods, Inheritance, Exception, Runtime Polymorphism, User defined Exception handling mechanism							
3	Illustrate inheritance, Exception handling mechanism, JDBC connectivity							
4	Construct Threads, Event Handling, implement packages							
Course Outcomes:								
S.No	Outcome							Knowledge Level
1	Construct Basic programs using object-oriented programming principles such as classes, objects, constructors, and polymorphism.							K3
2	Construct programs that utilize arrays, collections, and strings to solve a variety of computational problems.							K3
3	Apply the concepts of Inheritance, Packages to develop reusable programs.							K3
4	Develop robust programs using exception handling, multithreading and File I/O.							K3
5	Develop java application to interact with database by using relevant JDBC Driver.							K3
SYLLABUS								
1	Exercise-1 (Basics) 1. Write a JAVA program to display default value of all primitive data types. 2. Write a java program that display the roots of a quadratic equation $ax^2+bx=0$. Calculate the discriminate D and basing on value of D, describe the nature of root.							
2	Exercise- 2 (Classes & Constructors) 1. Write a JAVA program to implement class mechanism. Create a class, methods and invoke them inside main method. 2. Write a JAVA program to implement method overloading. 3. Write a JAVA program to implement constructor overloading. 4. Write a JAVA program to implement this keyword.							
3	Exercise– 3 (Arrays, Vectors, Strings) 1. Write a JAVA program to search for an element in a given list of elements using binary search mechanism. 2. Write a JAVA program to sort for an element in a given list of elements using bubble sort. 3. Write a JAVA program to implement Operations on Array list.							

	4. Write a JAVA program to implement Operations on Hash map. 5. Write a JAVA program to implement Operations on Hash set. 6. Write a JAVA program to implement String Operations. 7. Write a JAVA program to implement String Bufferclass.
4	Exercise – 4 (Inheritance & Interfaces) 1. Write a JAVA program to implement Single Inheritance. 2. Write a JAVA program to implement multilevel Inheritance. 3. Write a JAVA program for abstract class to find areas of different shapes. 4. Write a JAVA program that implements Runtime polymorphism (Method Overriding) 5. Write a JAVA program to implement “super” keyword. 6. Write a JAVA program to implement Interface.
5	Exercise-5 (Packages & Exception Handling) 1. Write a JAVA program to implement simple Packages. 2. Write a JAVA program to implement sub Packages. 3. Write a JAVA program to implement the following Builtin Exceptions. i. Arithmetic Exception. ii. Array Index Out of Bounds Exception iii. Number Format Exception. iv. Null Pointer Exception. 4. Write a JAVA program Illustrating Multiple catch clauses 5. Write a JAVA program to implement user defined Exception.
6	Exercise–6 (Multithreading & File IO) 1. Write a JAVA program that creates threads by extending Thread class .First thread display “Good Morning “every 1 sec, the second thread displays “Hello “every 2 seconds and the third display “Welcome” every 3 seconds. 2. Write a JAVA program to implement Runnable Interface. 3. Write a program to implement priorities to Thread. 4. Write a JAVA program to implement Thread Synchronization (Multiplication tables) 5. Write a JAVA program to copy contents of file into another using Byte Oriented IO. 6. Write a JAVA program to copy contents of file into another using Character Oriented IO.
7	Exercise-7 (JDBC) 1. Write a JDBC program to insert data into database. 2. Write a JDBC program to delete data from database. 3. Write a JDBC program to update data into database. 4. Write a JDBC program to retrieve data from database.
Reference Books:	
1	The complete Reference Java, 12th edition (2022), Herbert Schildt, Publisher: McGraw Hill.
2	Introduction to java programming, 9th edition (2014) by Y Daniel Liang, Publisher: Pearson.
3	Murach's Java Programming, 5th edition (2017) Joel Murach, Publisher: Mike Murach.
4	JdbcApi Tutorial and Reference 3E (2003), by Maydene, Jon Ellis (Author), Jonathan Bruce (Author), Publisher: Addison-Wesley Professional.

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CS2106	SEC	--	1	2	2	30	70	3 Hrs.
PYTHON PROGRAMMING								
(Common to CSE, CSG, CSIT, AIML & CIC Branches)								
Course Objectives:								
1	Introduce core programming concepts of Python programming language.							
2	Demonstrate about Python data structures like Lists, Tuples, Sets and dictionaries.							
3	Implement Functions, Modules in Python Programming and to create practical and contemporary applications using these.							
Course Outcomes:								
S.No	Outcome							Knowledge Level
1	Demonstrate various operations on Built-in data types & Strings							K3
2	Solve Computational Problems using Modular Programming through Functions and Modules							K3
3	Apply Data Science methods and techniques on real time data.							K3
SYLLABUS								
1	Introduction: Introduction to python programming, Installing and Using Jupyter Notebook. Parts of Python Programming Language: Identifiers, Keywords, Statements and Expressions, Variables, Operators, Precedence and Associativity, Data Types, Indentation, Comments, Reading Input, Print Output, Type Conversions, Dynamic and Strongly Typed Language. Control Flow Statements: if statement, if-else statement, if...elif...else, Nested if statement, while Loop, for Loop, continue and break Statements. Sample Experiments: 1. Write a program to find the largest element among three Numbers. 2. Write a Program to display all prime numbers within an interval 3. Write a program to swap two numbers without using a temporary variable. 4. Demonstrate the following Operators in Python with suitable examples. i. Arithmetic Operators ii. Relational Operators iii. Assignment Operators iv. Logical Operators v. Bit wise Operators vi. Ternary Operator vii. Membership Operators viii. Identity Operators 5. Write a program to add and multiply complex numbers 6. Write a program to print multiplication table of a given number.							

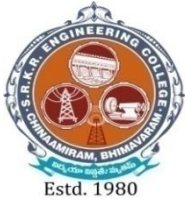
2	Functions & Modules: Built-In Functions, Commonly Used Modules, Function Definition and Calling the function, return Statement and void Function, Scope and Lifetime of Variables, Default Parameters, Keyword Arguments, *args and **kwargs, Command Line Arguments. Modules: random module, os module. Strings: Creating and Storing Strings, Basic String Operations, Accessing Characters in String by Index Number, String Slicing and Joining, String Methods, Formatting Strings. Lists: Creating Lists, Basic List Operations, Indexing and Slicing in Lists, Built-In Functions Used on Lists, List Methods, del Statement. Sample Experiments: 1. Write a program to define a function with multiple return values. 2. Write a program to define a function using default arguments. 3. Write a program to find the length of the string without using any library functions. 4. Write a program to check if the substring is present in a given string or not. 5. Write a program to perform the given operations on a list: i. addition ii. insertion iii. slicing 6. Write a program to perform any 5 built-in functions by taking any list.
3	Dictionaries: Creating Dictionary, Accessing and Modifying key: value Pairs in Dictionaries, Built-In Functions Used on Dictionaries, Dictionary Methods, del Statement. Tuples and Sets: Creating Tuples, Basic Tuple Operations, tuple() Function, Indexing and Slicing in Tuples, Built-In Functions Used on Tuples, Relation between Tuples and Lists, Relation between Tuples and Dictionaries, Using zip() Function, Sets, Set Methods, Frozenset. Sample Experiments: 1. Write a program to create tuples (name, age, address, college) for at least two members and concatenate the tuples and print the concatenated tuples. 2. Write a program to count the number of vowels in a string (No control flow allowed). 3. Write a program to check if a given key exists in a dictionary or not. 4. Write a program to add a new key-value pair to an existing dictionary. 5. Write a program to sum all the items in a given dictionary.
4	Files: Types of Files, Creating and Reading Text Data, File Methods to Read and Write Data, Reading and Writing Binary Files, Reading and Writing CSV Files Object-Oriented Programming: Classes and Objects, Creating Classes in Python, Creating Objects in Python, Constructor Method, Classes with Multiple Objects, Class Attributes Vs Data Attributes, Encapsulation, Inheritance, Polymorphism. Sample Experiments: 1. Write a program to sort words in a file and put them in another file. The output file should have only lower-case words, so any upper-case words from source must be lowered. 2. Python program to print each line of a file in reverse order. 3. Python program to compute the number of characters, words and lines in a file. 4. Write a program to create, display, append, insert and reverse the order of the items in the array.

	- Write a program to add, transpose and multiply two matrices. - Write a Python program to create a class that represents a shape. Include methods to calculate its area and perimeter. Implement subclasses for different shapes like circle, triangle, and square.
5	Introduction to Data Science: NumPy: NumPy arrays using Array function, Integer Indexing, Array Indexing, Boolean Array Indexing, Slicing and Iterating Arrays, Arithmetic Operations on NumPy arrays, Mathematical Functions in NumPy, changing shape of an array, Stacking and Splitting of arrays, Broad Casting in arrays. Pandas: Pandas Series, Pandas Data Frame Sample Experiments: - Python Program to demonstrate creation of Numpy Arrays. - Python program to demonstrate basic slicing, integer and Boolean indexing. - Python Program to Manipulate Numpy Arrays. - Python program to demonstrate Mathematical and Statistical Operations on Numpy Arrays. - Python program to create Series and Data Frame Objects using the sample data. - Python Program to demonstrate various operation on Series and Data Frame Objects.
Reference Books:	
1	Gowrishankar S, Veena A., Introduction to Python Programming, CRC Press.
2	Python Programming, S Sridhar, J Indumathi, V M Hariharan, 2 nd Edition, Pearson, 2024
3	Introduction to Programming Using Python, Y. Daniel Liang, Pearson
E-Resources	
1.	Programming in Python Coursera
2.	Learn Python - Free Interactive Python Tutorial

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23MC2101	MC	2	--	--	--	30	--	--
ENGLISH PROFICIENCY								
(Common to ME, IT, CSIT, AIDS& AIML)								
Course Outcomes: After completion of this course, students will be able to								
S.No	Outcome							Knowledge Level
1.	Interpret IELTS & TOEFL listening comprehension texts.							K2
2.	Demonstrate essential speaking skills in academic, professional, and real-life contexts.							K2
3.	Interpret the written discourse by applying effective reading strategies.							K2
4.	Construct coherent and cohesive paragraphs, e-mails, letters, and essays.							K3
SYLLABUS								
UNIT-I	Listening Skills Listening as a receptive skill Listening skills for IELTS, TOEFL, Duolingo, PTE, and other competitive examinations.							
UNIT-II	Speaking Skills JAM/ Extempore Debate / Group Discussion Presentation Skills							
UNIT-III	Reading Skills Types of Reading (Intensive and Extensive Reading, Skimming, Scanning) Reading/Summarizing News Paper Articles							
UNIT-IV	Writing Skills Essay Writing (Argumentative, Analytical and Descriptive essays for GRE, TOEFL & IELTS) E-Mail / Letter Writing Resume Writing							
UNIT-V	Integrated Language Skills Listening Skills for Speaking and Writing Reading Skills for Writing and Speaking							
Text Books:								
1.	Interchange (5 th edition) by Jack C.Richards, CUP, 2017.							
Reference Books:								
1.	Fundamentals of Technical Communication (1 st edition) by Meenakshi Raman, Sangeeta Sharma							

	of OUP, 2014.
2.	The Oxford Guide to Writing and Speaking (3 rd edition) by John Seely OUP, 2013.
3.	Effective Technical Communication (2 nd edition) by M. Ashraf Rizwi. TataMcGrawhill, 2017.
e-Resources:	
1	BBC Learning English - Learn English with BBC Learning English - Homepage
2	Grammar Learn English (britishcouncil.org)
3	Duolingo English Test
4	IELTS Test Preparation Materials - Videos, Practice tests, Articles and More (idp.com)





**SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE
(AUTONOMOUS)**

(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)

Accredited by NAAC with 'A+' Grade.

Recognised as Scientific and Industrial Research Organisation

SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., INDIA

Regulation: R23		II / IV - B.Tech. II - Semester							
ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING									
COURSE STRUCTURE (With effect from 2023-24 admitted Batch onwards)									
Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
B23HS2203	Optimization Techniques	HS	2	0	0	2	30	70	100
B23BS2201	Probability & Statistics	ES	3	0	0	3	30	70	100
B23AM2201	Machine Learning	PC	3	0	0	3	30	70	100
B23CS2202	Advanced Data Structures & Algorithm Analysis	PC	3	0	0	3	30	70	100
B23AM2202	Digital Logic & Computer Organization	PC	3	0	0	3	30	70	100
B23AM2203	Artificial Intelligence and Machine Learning Lab	PC	0	0	3	1.5	30	70	100
B23CS2205	Advanced Data Structures & Algorithm Analysis Lab	PC	0	0	3	1.5	30	70	100
B23AM2204	Full Stack Development-1	SEC	0	1	2	2	30	70	100
B23AM2205	Design Thinking & Innovation	ES	1	0	2	2	30	70	100
B23MC2202	Environmental Science	MC	2	0	0	-	30	-	30
TOTAL			17	1	10	21	300	630	930

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23HS2203	HS	2	--	--	2	30	70	3 Hrs.
OPTIMIZATION TECHNIQUES								
(Common to AIML & IT)								
Course Objectives:								
1.	To define an objective function and constraint functions in terms of design variables, and then state the optimization problem.							
2.	To state single variable and multi variable optimization problems, without and with constraints and solve by classical methods.							
3.	To apply linear programming technique to an optimization problem, define slack and surplus variables, by using Simplex method.							
4.	To apply transportation technique to real life problems of transportation.							
5.	To solve Integer programming problems that encounter in real life practical cases.							
6.	To design optimum schedules for projects.							
Course Outcomes: At the end of the course students will be able to								
S.No	Outcome							Knowledge Level
1.	Apply classical optimization techniques to minimize or maximize single and multi-variable optimization problems without or with constraints and arrive at an optimal solution.							K3
2.	Apply suitable optimization algorithms to formulate and solve linear programming problems.							K3
3.	Apply specialized methods to determine optimal schedules for transportation problems.							K3
4.	Solve Integer programming problems by using suitable optimization algorithms.							K3
5.	Apply analytical methods to project networks to determine optimal time schedules.							K3
SYLLABUS								
UNIT-I (10Hrs)	Classical Optimization Techniques: Statement of an Optimization problem, classification of Optimization problems, Single variable optimization without and with constraints, multi variable optimization without constraints, multivariable optimization with equality constraints - solution by the method of Lagrange multipliers, multivariable optimization with inequality constraints – Kuhn Tucker conditions.							
UNIT-II (10 Hrs)	Linear Programming: Definition and formulation of LPP, Graphical solution, Standard form of a linear							

	programming problem, Simplex method and Algorithm, Artificial Variable Technique – Big M method.
UNIT-III (10 Hrs)	Transportation Problem: Mathematical Formulation, Tabular Representation, Balanced and unbalanced transportation Problems - Initial Solution by north – west corner rule, least cost method and Vogel’s approximation method, testing for optimality by MODI method.
UNIT-IV (10 Hrs)	Integer Programming: Introduction to Integer programming, methods of integer programming, Gomory’s cutting plane method - All Integer Programming Problems, Branch and Bound method for all integer programming problems.
UNIT-V (10 Hrs)	Network Analysis: Introduction, Project Scheduling by CPM and PERT, Network diagram representation, rules for drawing network diagram, Labelling by Fulkerson’s rule, Network calculations - EST, EFT, LST, LFT, Float/Slack and critical path, CPM and PERT calculations.
Textbooks:	
1.	“Engineering optimization: Theory and practice”, S. S.Rao, New Age International (P) Limited, 3rd edition, 1998.
2.	Operations Research, Dr. S. D. Sharma, Kedarnath, Ramnath & Co
Reference Books:	
1.	“Optimization Methods in Operations Research and systems Analysis”, by K.V. Mital and C. Mohan, New Age International (P) Limited, Publishers, 3rd edition, 1996.
2.	“Introductory Operations Research”, H.S. Kasene & K.D. Kumar, Springer (India), Pvt.LTd
e-Resources	
1.	https://nptel.ac.in/courses/112/106/112106134/
2.	https://nptel.ac.in/courses/110/106/110106062/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23BS2201	ES	3	0	0	3	30	70	3 Hrs.
PROBABILITY AND STATISTICS								
(Common to AIML, CSE, CSBS, CSIT and IT)								
Course Objectives: Students are expected to								
1.	Acquire critical thinking skills from the concepts of Descriptive Statistics, Data Science and Probability.							
2.	Predict the relationship and the impact of the relationship between the variables.							
3.	Obtain Decision-making skills from Sampling theory.							
Course Outcomes: Upon the successful completion of this course, Students will								
S.No.	Outcome							Knowledge Level
1.	Compute various statistical measures like central tendency and spread values.							K2
2.	Use the concepts of probability and random variables to solve simple problems based on discrete and continuous probability distributions.							K3
3.	Determine correlation and regression coefficients and model a best suitable curve for a given data using the method of least squares.							K3
4.	Apply the procedures of sampling theory to find point and interval estimates for various sampling distributions.							K3
5.	Model a framework by testing of hypothesis for getting inferences about Population Parameters based on Sample statistic.							K3
SYLLABUS								
UNIT-I (10 Hrs.)	Descriptive statistics and methods for data science: Data science- Statistics Introduction- Population vs Sample –Collection of data: primary and secondary data- Type of variables: dependent and independent, Categorical and Continuous variables- Data visualization- Measures of Central tendency- Measures of Variability (spread or variance)- Moments- Measures of Skewness and Kurtosis.							
UNIT-II (12 Hrs.)	Probability, Random variables and Distributions: Probability– Conditional probability, addition, Multiplication theorems and Baye’s theorem. Random variables – Discrete and Continuous random variables – Distribution functions – Probability mass function, Probability density function and Cumulative distribution functions – Mathematical Expectation – Variance. Discrete and Continuous Distributions: Discrete Distributions: Bernoulli, Binomial and Poisson distributions - Mean, Variance, Fitting of distributions. Continuous Distributions: Uniform distribution, Normal Distribution, Standard Normal Variate - Mean, Variance.							

UNIT-III (12 Hrs.)	Correlation, Regression and Curve fitting: Correlation: Definition, Karl Pearson's Coefficient of Correlation, Limits for correlation coefficient, Rank Correlation, Spearman's rank correlation coefficient (without proofs). Linear Regression: Regression- Regression coefficients and properties–Multiple Linear Regression. Curve Fitting: Method of Least Squares, Fitting of a Straight line, Fitting of a Parabola, Fitting of an Exponential curves: $y = ae^{bx}$, $y = ab^x$ and Power curve: $y = ax^b$.
UNIT-IV (8 Hrs.)	Sampling Theory: Introduction – Population and Samples –Parameter and Statistic- Sampling distribution of statistic-Standard error- Sampling distribution of Means and Variance (definition only) – Point and Interval estimations. Maximum error of estimate – Central limit theorem (without proof) – Estimation using t, χ^2 and F-distributions.
UNIT-V (12 Hrs.)	Testing of Hypothesis: Testing of Hypothesis- Formulation of Null hypothesis, Alternative hypothesis, Critical region, level of significance, Errors in sampling- Type-I-error, Type-II-error, One-tailed and Two-tailed tests, Degrees of freedom. Large Sample Theory: Test of significance for single and difference of Proportions. Small Sample Theory: Student's-t-distribution: Definition, t-test for single mean, t-test for difference of means, Paired t-test for difference of means. F-distribution: Definition, F-test for equality of two population variances. Chi-square distribution: Definition, Chi-square test for attributes and goodness of fit.
Text Books:	
1.	Fundamentals of Mathematical Statistics by S. C. Gupta and V.K. Kapoor, Sultan Chand & Sons Publishers.
2	Probability and Statistics for Engineers, Miller and Freund, 7 th edition, Prentice-Hall India.
Reference Books:	
1.	Probability and statistics for Engineers and Scientists by Ronald E. Walpole, Raymond H. Myers, Sharon L. Myers and Keying Ye, Eighth edition, Pearson Education.
2.	Probability and Statistics for Engineering and the Sciences, Jay I. Devore, 8 th Edition, Cengage.
3.	Introduction to probability and statistics Engineers and the Scientists, Sheldon M. Ross, 4 th Edition, Academic Foundation, 2011.
4.	Johannes Ledolter and Robert V. Hogg, Applied statistics for Engineers and Physical Scientists, 3rd Edition, Pearson, 2010.
5.	Probability, Statistics and Random Processes by T.Veerarajan, Tata McGraw Hill Pub.
6.	Higher Engineering Mathematics, by Dr.B.S.Grewal, 43 rd Edition, Khanna Publishers
7.	Paul L. Meyer, Introductory Probability and Statistical Applications (2 nd edn.), Addison-Wesley, 1970.
e-Resources:	
1.	http://www.swayam.gov.in
2.	https://archive.nptel.ac.in/courses/106/104/106104233/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23AM2201	PC	3	--	--	3	30	70	3 Hrs.
MACHINE LEARNING								
(For AIML)								
Course Objectives:								
1.	Define machine learning and its different types (supervised and unsupervised) and understand their applications.							
2.	Understand data representation, feature selection and feature extraction.							
3.	Apply supervised learning algorithms including decision trees and k-nearest neighbors (k-NN), Linear Discriminants.							
4.	Implement unsupervised learning techniques, such as K-means clustering, Hierarchical Clustering, Fuzzy C-means clustering, Expectation Maximization Clustering.							
Course Outcomes: After the completion of the course students will be able to								
S. No.	Outcome							Knowledge Level
1.	Apply different Machine Learning paradigms, project development stages and Feature engineering							K3
2.	Use proximity measures, distance-based classification and regression algorithms, model evaluation metrics on given data							K3
3.	Solve classification and regression problems using decision tree algorithms and Bayes Classifier							K3
4.	Apply linear discriminants to solve classification problems							K3
5.	Find clusters in the given dataset using partitional, hierarchical, fuzzy, probabilistic and graph-based clustering algorithms							K3
SYLLABUS								
UNIT-I (10Hrs)	Introduction to Machine Learning: Evolution of Machine Learning, Paradigms for ML, Learning by Rote, Learning by Induction, Reinforcement Learning, Types of Data, Matching, Stages in Machine Learning, Data Acquisition, Feature Engineering, Data Representation, Model Selection, Model Learning, Model Evaluation, Model Prediction, Search and Learning, Data Sets. Representation: Introduction to Representation, Feature selection, Linear Feature Extraction, Eigen Values and Eigen Vectors, PCA, SVD.							
UNIT-II (10 Hrs)	Nearest Neighbor-Based Models: Introduction to Proximity Measures, Distance Measures, Non-Metric Similarity Functions, Proximity Between Binary Patterns, Different Classification Algorithms Based on the Distance Measures, K-Nearest Neighbor Classifier, Radius Distance Nearest Neighbor Algorithm, KNN Regression, Performance of Classifiers, Performance of Regression Algorithms.							

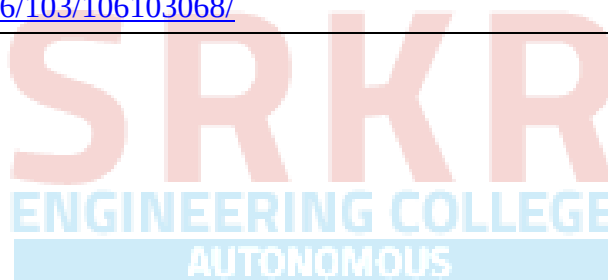
UNIT-III (10 Hrs)	Models Based on Decision Trees: Decision Trees for Classification, Impurity Measures, Properties, Regression Based on Decision Trees, Bias–Variance Trade-off, Random Forests for Classification and Regression. The Bayes Classifier: Introduction to the Bayes Classifier, Bayes’ Rule and Inference, The Bayes Classifier and its Optimality, Multi-Class Classification Class Conditional Independence and Naive Bayes Classifier (NBC).
UNIT-IV (10 Hrs)	Linear Discriminants for Machine Learning: Introduction to Linear Discriminants, Linear Discriminants for Classification, Perceptron Classifier, Perceptron Learning Algorithm, Support Vector Machines, Linearly Non-Separable Case, Non-linear SVM, Kernel Trick, Logistic Regression, Linear Regression, Multi-Layer Perceptrons (MLPs), Backpropagation for Training an MLP.
UNIT-V (10 Hrs)	Clustering: Introduction to Clustering, Partitioning of Data, Matrix Factorization Clustering of Patterns, Divisive Clustering, Agglomerative Clustering, Partitional Clustering, K-Means Clustering, Soft Partitioning, Soft Clustering, Fuzzy C-Means Clustering, Rough Clustering, Rough K-Means Clustering Algorithm, Expectation Maximization-Based Clustering, Spectral Clustering.
Textbooks:	
1.	“Machine Learning Theory and Practice”, M N Murthy, V S Ananthanarayana, Universities Press (India), 1 st Edition, 2024.
Reference Books:	
1.	“Machine Learning”, Tom M. Mitchell, 1 st Edition, McGraw-Hill Publication, 2017.
2.	“Machine Learning in Action”, 1 st Edition, Peter Harrington, DreamTech, 2012
3.	“Introduction to Data Mining”, Pang-Ning Tan, Michel Stenbach, Vipin Kumar, 2 nd Edition, 2021.
e-Resources	
1.	https://onlinecourses.nptel.ac.in/noc24_cs81/preview
2.	https://onlinecourses.nptel.ac.in/noc24_cs51/preview
3.	https://www.coursera.org/specializations/machine-learning-introduction

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CS2202	PC	3	--	-	3	30	70	3 Hrs.
ADVANCED DATA STRUCTURES & ALGORITHM ANALYSIS								
(Common to CSE, CSIT, AIML & CSG Branches)								
Course Objectives:								
1.	Provide knowledge on advanced data structures frequently used in Computer Science domain.							
2.	Develop skills in algorithm design techniques popularly used.							
3.	Understand the use of various data structures in the algorithm design.							
Course Outcomes: At the end of the course Students will be able to,								
S.No	Outcome							Knowledge Level
1.	Use advanced data structures to organize data and solve connectivity problems.							K3
2.	Analyze the time complexity of Divide and Conquer based algorithms.							K4
3.	Apply Greedy and Backtracking strategies to solve computational problems.							K3
4.	Use Dynamic programming strategy to solve optimization problems.							K3
5.	Determine solutions for combinatorial optimization problems.							K3
SYLLABUS								
UNIT-I (12Hrs)	Trees : AVL Trees – Creation, Insertion, Deletion operations, Red-Black Trees – Creation, Insertion, Deletion operations, Heap Trees (Priority Queues) – Min and Max Heaps, Operations and Applications. Graphs: Terminology, Representations, Basic Search and Traversals, Connected and Bi-connected Components, Euler circuits.							
UNIT-II (08 Hrs)	Introduction to Algorithm Analysis, Space and Time Complexity analysis, Asymptotic Notations. Divide and Conquer: The General Method, Quick Sort, Merge Sort, Strassen’s matrix multiplication, Convex Hull, Time complexity analysis of divide and conquer algorithms.							
UNIT-III (10 Hrs)	Greedy Method: General Method, Job Sequencing with deadlines, Knapsack Problem, Minimum cost spanning trees, Single Source Shortest Paths. Backtracking: General Method, 8-Queens Problem, Sum of Subsets problem, Graph Coloring, 0/1 Knapsack Problem.							
UNIT-IV (10 Hrs)	Dynamic Programming: General Method, All pairs shortest paths, Single Source Shortest Paths – General Weights (Bellman Ford Algorithm), Optimal Binary Search							

	Trees, 0/1 Knapsack, String Editing, Traveling Salesperson problem.
UNIT-V (10 Hrs)	Branch and Bound: The General Method, 0/1 Knapsack Problem, Travelling Salesperson problem. NP Hard and NP Complete Problems: Basic Concepts, Cook's theorem. NP Hard Graph Problems: Clique Decision Problem (CDP), Chromatic Number Decision Problem (CNDP), Traveling Salesperson Decision Problem (TSP).
Textbooks:	
1.	Data Structures and Algorithm Analysis in C, Mark Allen Weiss, 2nd Edition ,Pearson Edu Publishers, 2007.
2.	Computer Algorithms/C++ Ellis Horowitz, Sartaj Sahni, Sanguthevar Rajasekaran , 2nd Edition University Press,2008.
Reference Books:	
1.	Fundamentals of Data Structures in C++, Horowitz Ellis, SahniSartaj, Mehta, Dinesh, 2nd Edition, Universities Press, 2008.
2.	Data Structures and Algorithms: Concepts, Techniques and Applications – G.A.V.Pai, 1st Edition TataMcGraw Hill Publishers,2017.
3.	Data Structures and program design in C, Robert Kruse, 2 nd Edition, ,Pearson Education Asia,2006.
4.	An introduction to Data Structures with applications, Trembley& Sorenson,2nd Edition ,McGraw Hill,2010.
5.	The Art of Computer Programming, Vol.1: Fundamental Algorithms, Donald E Knuth, Addison- Wesley, 1997.
e-Resources	
1.	https://www.tutorialspoint.com/advanced_data_structures/index.asp
2.	http://peterindia.net/Algorithms.html

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23AM2202	PC	3	--	--	3	30	70	3 Hrs.
DIGITAL LOGIC & COMPUTER ORGANIZATION								
(For AIML)								
Course Objectives: The main objectives of the course is to provide students with:								
1.	A comprehensive understanding of digital logic design principles and computer organization fundamentals.							
2.	Describe memory hierarchy concepts.							
3.	Explain input/output (I/O) systems and their interaction with the CPU.							
4.	Memory and peripheral devices.							
Course Outcomes: At the end of the course Students will be able to								
S.No	Outcome							Knowledge Level
1.	Use various logic gates to design combinational and sequential circuits							K3
2.	Use computer arithmetic operations on Fixed and floating point numbers							K3
3.	Describe the functional unit of Central Processing Unit							K2
4.	Explain various types of memory organizations							K2
5.	Explain Input/output organizations							K2
SYLLABUS								
UNIT-I (10 Hrs)	Fundamentals of Logic gate: Basic Logic Functions, Logic gates, universal logic gates, Minimization of Logic expressions, K-Map Simplifications, Don't Cares. Combinational Circuits: Decoders, Multiplexers, and Encoders. Sequential Circuits: Flip-Flops, Flip-Flop Conversions, Shift Registers and Ripple counters.							
UNIT-II (12 Hrs)	Basic Structure of Computers: Computer Types, Functional units, Basic operational concepts, Bus structures, multiprocessors and multi computers, Computer Generations, Von-Neumann Architecture. Computer Arithmetic: Fixed and floating-point representation of numbers, Addition, Subtraction, Multiplication and Division algorithms, Floating- point arithmetic operations.							
UNIT-III (12 Hrs)	Central Processing Unit: Introduction, General Register Organization, Stack Organization, Instruction Formats, Addressing Modes, Data Transfer and Manipulation Hardwired Control and Multi programmed Control: Control Memory, Address Sequencing, Micro program Example, Design of Control Unit.							

UNIT-IV (08 Hrs)	Memory Organization: Memory Hierarchy, Secondary Storage Memory, Associative Memory, Cache Memory, Virtual memory.
UNIT-V (08 Hrs)	Input/output Organization: Peripheral devices, I/O interface, Asynchronous data transfer, Modes of transfer, Priority interrupt, direct memory access and IOP.
Textbooks:	
1.	Digital Design, 6th Edition, M. Morris Mano, Pearson Education.
2.	Computer Systems Architecture, M.Moris Mano, 3rdEdition, Pearson.
3.	Computer Organization, Carl Hamacher, Zvonko Vranesic, Safwat Zaky, 6th edition, McGraw Hill.
Reference Books:	
1.	Computer Organization and Architecture, William Stallings, 11thEdition, Pearson.
2.	Computer Organization and Design, David A. Paterson, John L. Hennessy, Elsevier.
3.	Fundamentals of Logic Design, Roth, 5thEdition, Thomson.
e-Resources:	
1.	https://nptel.ac.in/courses/106/103/106103068/



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23AM2203	PC	--	--	3	1.5	30	70	3 Hrs.
ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING LAB								
(For AIML)								
Course Objectives:								
1	The student should be made to learn the problem-solving methods using Artificial Intelligence.							
2	To learn about different data preprocessing techniques.							
3	To learn about classification and regression algorithms.							
4	To apply different clustering algorithms on different datasets.							
Course Outcomes: After the completion of the course students will be able to								
S. No.	Outcome							Knowledge Level
1	Apply Heuristic Search techniques to solve AI Problems							K3
2	Select appropriate data preprocessing techniques for the given dataset							K5
3	Evaluate classification and clustering techniques based on performance							K5
4.	Apply machine learning concepts and algorithms to develop a mini project addressing a practical real-world problem.							K3
SYLLABUS								
1	Implement Breadth First Search using Python.							
2	Implement Depth First Search using Python.							
3	Implement A* algorithm using Python. (Ex: find the shortest path)							
4	Implement AO* algorithm using Python (Ex: find the shortest path)							
5	Apply the following Pre-processing techniques on real-life datasets a. Attribute selection b. Handling Missing Values c. Discretization d. Elimination of Outliers							
6	Demonstrate dimensionality reduction using PCA.							
7	Apply KNN algorithm for classification and regression.							
8	Demonstrate decision tree algorithm for a classification problem and perform parameter tuning for better results.							
9	Apply Random Forest algorithm for classification and regression.							
10	Demonstrate Naïve Bayes Classification algorithm.							
11	Apply Support Vector algorithm for classification.							
12	Demonstrate Linear Regression and Logistic Regression models.							
13	Apply Lasso and Ridge Regularization to reduce the variance of a linear regression model							

14	Implement the K-means algorithm and apply it to the data you selected. Evaluate performance by measuring the sum of the Euclidean distance of each example from its class center. Test the performance of the algorithm as a function of the parameters .
15.	Develop a mini project that applies machine learning concepts and algorithms to address a practical real-world problem.

Reference Books:

1.	Chris Albon, “Machine Learning with Python Cookbook-practical solutions from preprocessing to Deep learning”, O’REILLY Publisher, 2018
2.	Sebastian Raschka & Vahid Mirjalili, “Python Machine Learning”, Packt Publisher, 2017
3.	AI and Machine Learning for Coders: A Programmer's Guide to Artificial Intelligence, Indian Edition, Laurence Moroney, 2020

e-Resources:

1.	https://data.world/datasets/machine-learning
2.	https://www.ml4devs.com/articles/datasets-for-machine-learning-and-data-science/
3.	https://www.kaggle.com/datasets
4.	https://archive.ics.uci.edu/



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CS2205	PC	--	--	3	1.5	30	70	3 Hrs.
ADVANCED DATA STRUCTURES & ALGORITHM ANALYSIS LAB								
(Common to CSE, CSIT, AIML & CSG Branches)								
Course Objectives:								
1	Acquire practical skills in constructing and managing Data structures.							
2	Apply the popular algorithm design methods in problem-solving scenarios.							
Course Outcomes: At the end of the course student will be able to								
S.No	Outcome							Knowledge Level
1	Demonstrate the basic operations of advanced trees.							K3
2	Apply graph traversal techniques to solve connectivity problems.							K3
3	Analyze the performance of sorting algorithms for various inputs.							K4
4	Apply different algorithm strategies to find solutions for various problems.							K3
SYLLABUS								
1	Construct an AVL tree for a given set of elements which are stored in a file. and implement insert and delete operation on the constructed tree. Write contents of tree into a new file using in-order.							
2	Construct Min and Max Heap using arrays, delete any element and display the content of the Heap.							
3	Implement BFT and DFT for given graph, when graph is represented by a) Adjacency Matrix b) Adjacency Lists							
4	Write a program for finding the bi-connected components in a given graph.							
5	Implement Quick sort and Merge sort and observe the execution time for various input sizes (Average, Worst and Best cases).							
6	Compare the performance of Single Source Shortest Paths using Greedy method when the graph is represented by adjacency matrix and adjacency lists.							
7	Implement Job sequencing with deadlines using Greedy strategy.							
8	Implement N-Queens Problem Using Backtracking.							
9	Use Backtracking strategy to solve 0/1 Knapsack problem.							
10	Write a program to solve 0/1 Knapsack problem Using Dynamic Programming.							
11	Implement Travelling Sales Person problem using Branch and Bound approach.							
Reference Books								
1	Data Structures and Algorithm Analysis in C, Mark Allen Weiss, 2nd Edition, Pearson Edu Publishers,2002							
2	Computer Algorithms/C++ Ellis Horowitz, Sartaj Sahni, Sanguthevar Rajasekaran 2nd Edition							

	University Press,2008
3	Data Structures and program design in C, Robert Kruse, 2nd Edition, Pearson Education Asia,2006.
E-Resources	
1.	https://www.tutorialspoint.com/advanced_data_structures/index.asp
2.	http://peterindia.net/Algorithms.html
3.	https://github.com/GDSC-KIIT/DSA-Resource?stab=readme-ov-file
4.	https://www.geeksforgeeks.org/design-and-analysis-of-algorithms/?ref=lbp
5.	https://www.cs.usfca.edu/~galles/visualization/Algorithms.html



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23AM2204	SEC	0	1	2	2	30	70	3 Hrs.
FULL STACK DEVELOPMENT – 1								
(Common to CSE, CSIT, CIC, AIML & CSG Branches)								
Course Objectives: The main objectives of the course are to								
1	Make use of HTML elements and their attributes for designing static web pages							
2	Build a web page by applying appropriate CSS styles to HTML elements							
3	Experiment with Java Script to develop dynamic web pages and validate forms							
Course Outcomes: At the end of the course students will be able to								
S.No	Outcome							Knowledge Level
1	Use HTML to implement static web pages							K3
2	Apply Cascading style sheets to style HTML documents							K3
3	Use Java Script to implement dynamic web pages							K3
SYLLABUS								
1	HTML Lists, Links and Images a) Write a HTML program, to explain the working of lists. Note: It should have an ordered list, unordered list, nested lists and ordered list in an unordered list and definition lists. b) Write a HTML program, to explain the working of hyperlinks using<a> tag and href, target attributes. c) Create a HTML document that has your image and your friend’s image with a specific height and width. Also when clicked on the images it should navigate to their respective profiles. d) Write a HTML program, in such a way that, rather than placing large images on a page, the preferred technique is to use thumbnails by setting the height and width parameters to something like to 100*100 pixels. Each thumbnail image is also a link to a full sized version of the image. Create an image gallery using this technique							
2	HTML Tables and Forms a) Write a HTML program, to explain the working of tables. (usetags: <table>, <tr>, <th>, <td> and attributes: border, row span, col span) b) Write a HTML program, to explain the working of tables by preparing a timetable. (Note: Use <caption> tag to set the caption to the table & also use cellspacing, cellpadding, border, rowspan, colspan etc.). c) Write a HTML program, to explain the working of forms by designing Registration form.(Note: Include text field, password field, number field, date of birth field, check boxes, radio buttons, list boxes using <select>&<option> tags, <text area> and two buttons ie: submit and reset. Use tables to provide a better view).							

	d) Write a HTML program, that makes use of <article>, <aside>, <figure>, <figcaption>, <footer>, <header>, <main>, <nav>, <section>, <div>, tags. e) Write a HTML program, to embed audio and video into HTML webpage.
3	Cascading Style Sheets, Selector forms Write a program to apply different types of selector forms - Simple selector (element, id, class, group, universal) - Combinator selector (descendant, child, adjacent sibling, general sibling) - Pseudo-class selector - Pseudo-element selector - Attribute selector
4	Types of CSS, CSS with Color, Background, Font, Text and CSS Box Model - Write a program to apply different types (or levels of styles or style specification formats) - inline, internal, external styles to HTML elements. (Identify selector, property and value). - Write a program to demonstrate the various ways you can reference a color in CSS. - Write a CSS rule that places a background image halfway down the page, tilting it horizontally. The image should remain in place when the user scrolls up or down. - Write a program using the following terms related to CSS font and text: - font-size - font-weight - font-style - text-decoration - text-transformation - text-alignment - Write a program, to explain the importance of CSS Box model using - Content - Border - Margin - padding
5	Applying JavaScript-internal and external, I/O, Type Conversion - Write a program to embed internal and external Java Script in a webpage. - Write a program to explain the different ways for taking input and displaying output. - Create a webpage which uses prompt dialogue box to ask a voter for his name and age. Display the information in table format along with either the voter can vote or not
6	JavaScript Pre-defined and User-defined Objects - Write a program using document object properties and methods. - Write a program using window object properties and methods. - Write a program using math object properties and methods. - Write a program using string object properties and methods. - Write a program using regex object properties and methods. - Write a program using date object properties and methods.
7	JavaScript Conditional Statements and Loops Write a program which asks the user to enter three integers, obtains the numbers from the user and outputs HTML text that displays the larger number followed by the words

	“LARGER NUMBER” in an information message dialog. If the numbers are equal, output HTML text as “EQUAL NUMBERS”. b) Write a program to display week days using switch case. c) Write a program to print 1 to 10 numbers using for, while and do-while loops. d) Write a program to print data in object using for-in, for-each and for-of loops. e) Develop a program to determine whether a given number is an ‘ARMSTRONG NUMBER’ or not. (Eg: 153 is an Armstrong number, since sum of the cube of the digits is equal to the number i.e., $1^3 + 5^3 + 3^3 = 153$). f) Write a program to display the denomination of the amount deposited in the bank in terms of 100’s, 50’s, 20’s, 10’s, 5’s, 2’s & 1’s. (Eg: If deposited amount is Rs.163, the output should be 1-100’s, 1-50’s, 1- 10’s, 1-2’s & 1- 1’s)
8	Java Script Functions and Events a) Design a HTML having a text box and four buttons named Factorial, Fibonacci, Prime, and Palindrome. When a button is pressed an appropriate function should be called to display - Factorial of that number - Fibonacci series up to that number - Prime numbers up to that number - Is it palindrome or not b) Write a program to validate the following fields in a registration page - Name (start with alphabet and followed by alphanumeric and the length should not be less than 6 characters) - Mobile(only numbers and length 10 digits) - E-mail (should contain form at like xxxxxxxx@xxxxxx.xxx)
9	Mini Project
Reference Books:	
1	Programming the World Wide Web, 7th Edition, Robert W Sebesta, Pearson, 2013.
2	Web Programming with HTML5, CSS and JavaScript, John Dean, Jones & Bartlett Learning, 2019 (Chapters 1-11).
3	Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node, Vasanth Subramanian, 2nd edition, APress, O’Reilly.
Web Links:	
1	https://www.w3schools.com/html
2	https://www.w3schools.com/css
3	https://www.w3schools.com/js/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23AM2205	ES	1	--	2	2	30	70	3 Hrs.
DESIGN THINKING & INNOVATION								
(Common to all Programmes of Engineering)								
Course Objectives:								
1.	Bring awareness on innovative design and new product development.							
2.	Explain the basics of design thinking.							
3.	Familiarize the role of reverse engineering in product development.							
4.	Train how to identify the needs of society and convert into demand.							
5.	Introduce product planning and product development process.							
Course Outcomes: After completion of this course, students will be able to								
S.No	Outcome							Knowledge Level
1.	Define the concepts related to design thinking.							K1
2.	Explain the fundamentals of Design Thinking and innovation.							K2
3.	Apply the design thinking techniques for solving problems in various sectors.							K3
4.	Analyse to work in a multidisciplinary environment.							K4
5.	Evaluate the value of creativity.							K5
SYLLABUS								
UNIT-I (10Hrs)	Introduction to elements and principles of Design, basics of design-dot, line, shape, form as fundamental design components. Principles of design. Introduction to design thinking, history of Design Thinking, New materials in Industry.							
UNIT-II (10 Hrs)	Design thinking process (empathize, analyze, idea & prototype), implementing the process in driving inventions, design thinking in social innovations. Tools of design thinking - person, costumer, journey map, brainstorming, product development. Activity: Every student presents their idea in three minutes, Every student can present design process in the form of flow diagram or flow chart etc. Every student should explain about product development.							
UNIT-III (10 Hrs)	Art of innovation, Difference between innovation and creativity, role of creativity and innovation in organizations. Creativity to Innovation. Teams for innovation, Measuring the impact and value of creativity. Activity: Debate on innovation and creativity, Flow and planning from idea to innovation, Debate on value-based innovation.							
UNIT-IV (10 Hrs)	Problem formation, introduction to product design, Product strategies, Product value, Product planning, product specifications. Innovation towards product design Case studies. Activity: Importance of modeling, how to set specifications, Explaining their own product							

	design.
UNIT-V (10 Hrs)	Design Thinking applied in Business & Strategic Innovation, Design Thinking principles that redefine business – Business challenges: Growth, Predictability, Change, Maintaining Relevance, Extreme competition, Standardization. Design thinking to meet corporate needs. Design thinking for Startups. Defining and testing Business Models and Business Cases. Developing & testing prototypes.
	Activity: How to market our own product, about maintenance, Reliability and plan for startup.
Textbooks:	
1.	Tim Brown, Change by design, 1/e, Harper Bollins, 2009.
2.	Idris Mootee, Design Thinking for Strategic Innovation, 1/e, Adams Media, 2014.
Reference Books:	
1.	David Lee, Design Thinking in the Classroom, Ulysses press, 2018.
2.	Shrrutin N Shetty, Design the Future, 1/e, Norton Press, 2018.
3.	William lidwell, Kritinaholden, &Jill butter, Universal principles of design, 2/e, Rockport Publishers, 2010.
4.	Chesbrough.H, The era of open innovation, 2003.
e-Resources:	
1.	https://nptel.ac.in/courses/110/106/110106124/
2.	https://nptel.ac.in/courses/109/104/109104109/
3.	https://swayam.gov.in/nd1_noc19_mg60/preview
4.	https://onlinecourses.nptel.ac.in/noc22_de16/preview

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23MC2202	MC	2	--	--	--	30	--	--
ENVIRONMENTAL SCIENCE								
(Common to AIDS, AIML, CE, CSBS, CSIT, IT and ME.)								
Course Objectives: The objective of the course is to impart:								
1.	Overall view on natural resources.							
2.	Awareness on ecosystem and its services.							
3.	Various environmental challenges induced due to unplanned anthropogenic activities.							
4.	Consciousness on the social issues, environmental legislation and global treaties							
Course Outcomes: At the end of the course, the students will be able to								
S. No	Outcome							Knowledge Level
1	Describe natural resources and their interaction							K2
2	Illustrate ecosystem types, biodiversity and conservation strategies							K2
3	Summarize contaminants of environment and preventive methods							K2
4	Explain protection of environment by employing constitutional provisions							K2
5	Explain global scenario of surroundings and social conditions							K2
SYLLABUS								
UNIT-I (8 Hrs)	Multidisciplinary Nature of Environmental Studies: Definition, Scope and Importance – Need for Public Awareness. Natural Resources: Renewable and non-renewable resources – Natural resources and associated problems. Forest resources – Use and over – exploitation, deforestation, case studies – Timber extraction – Mining, dams and other effects on forest and tribal people. Water resources – Use and over utilization of surface and ground water – Floods, drought, conflicts over water, dams – benefits and problems. Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies. Food resources: World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies. Energy resources: Growing energy needs, renewable and non-renewable energy sources use of alternate energy sources							
UNIT-II (8 Hrs)	Ecosystems: Concept of an ecosystem. – Structure and function of an ecosystem – Producers, consumers and decomposers – Energy flow in the ecosystem – Ecological succession – Food chains, food webs and ecological pyramids – Introduction, types, characteristic features, structure and function of the following ecosystem: a. Forest ecosystem. b. Grassland ecosystem c. Desert ecosystem d. Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries) Biodiversity and Its Conservation : Introduction and Definition - genetic, species and ecosystem diversity – Bio-geographical classification of India – Value of biodiversity:							

	consumptive use, Productive use, social, ethical, aesthetic and option values – Biodiversity at global, National and local levels – India as a mega-diversity nation – Hot-spots of biodiversity – Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts – Endangered and endemic species of India – Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity.
UNIT-III (6 Hrs)	Environmental Pollution: Definition, Cause, effects and control measures of: a. Air Pollution. b. Water pollution c. Soil pollution d. Marine pollution e. Noise pollution f. Thermal pollution g. Nuclear hazards. Solid Waste Management: Causes, effects and control measures of urban and industrial wastes – Role of an individual in prevention of pollution – Pollution case studies – Disaster management: floods, earthquake, cyclone and landslides
UNIT-IV (6 Hrs)	Social Issues and the Environment: From Unsustainable to Sustainable development – Urban problems related to energy watershed management – Resettlement and rehabilitation of people; its problems and concerns – Carbon credits, Mission LiFE. Environmental ethics: Issues and possible solutions. Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust –Wasteland reclamation – Consumerism and waste products. Environment Protection Act. – Air (Prevention and Control of Pollution) Act. – Water (Prevention and control of Pollution) Act – Wildlife Protection Act – Forest Conservation Act. Issues involved in enforcement of environmental legislation – Public awareness.
UNIT-V (6 Hrs)	Human Population And The Environment: Population growth, variation among nations, Population explosion – Family Welfare Programmes. – Environment and human health – Human Rights – Value Education – HIV/AIDS – Women and Child Welfare – Role of information Technology in Environment and human health – Case studies. Field Work: Visit to a local area to document environmental assets River/forest grassland/hill/mountain – Visit to a local polluted site- Urban/Rural/Industrial/Agricultural Study of common plants, insects, and birds – river, hill slopes, etc.
Text Books:	
1.	Erach Bharucha, Text book of Environmental Studies for Undergraduate Courses, Universities Press (India) Private Limited, 2019.
2.	Palaniswamy, Environmental Studies, 2/e, Pearson education, 2014.
3.	S. Azeem Unnisa, Environmental Studies, Academic Publishing Company, 2021.
4.	K. Raghavan Nambiar, “Text book of Environmental Studies for Undergraduate Courses as per UGC model syllabus”, SciTech Publications (India), Pvt. Ltd, 2010.
5	K. V. S. G. Murali Krishna, The Book of Environmental Studies, Savera Publishing House.
6	Environmental Studies, R. Rajagopalan, 2 nd Edition, 2011, Oxford University Press.
Reference Books:	

1.	Deeksha Dave and S.S. Katewa, Textbook of Environmental Studies, 2/e, Cengage Learning.
2.	M. Anji Reddy, “Textbook of Environmental Sciences and Technology”, BS Publication, 2014.
3.	J.P. Sharma, Comprehensive Environmental studies, Laxmi publications, 2006.
4.	J. Glynn Henry and Gary W. Heinke, Environmental Sciences and Engineering, Prentice Hall of India Private limited, 1988.
5.	G.R. Chatwal, A Text Book of Environmental Studies, Himalaya Publishing House, 2018
6.	Gilbert M. Masters and Wendell P. Ela, Introduction to Environmental Engineering and Science, 1/e, Prentice Hall of India Private limited, 1991.

e-Resources

1.	https://onlinecourses.nptel.ac.in/noc23_hs155/preview
2.	https://www.edx.org/learn/environmental-science/rice-university-ap-r-environmental-science-part-3-pollution-and-resources?index=product&objectID=course-3a6da9f2-lec07.pdf (iasri.res.in)
3.	http://ecoursesonline.iasri.res.in/Courses/Environmental%20Science-I/Data%20Files/pdf/lec07.pdf
4.	https://www.youtube.com/watch?v=5QxxaVfgQ3k

